

Course: B. Tech.

Branch : Mechanical Engineering

Semester : V

Subject Code & Name: BTMC 501 & Heat Transfer

Max Marks: 60

Date: 07/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

- A) Write a note on thermal conductivity. Give some examples of heat transfer in engineering. (L1/CO1) 6
- B) The inner surface of furnace wall is at 200 °C and outer surface is at 50 °C. Calculate the heat loss per m² area of the wall. If the thermal conductivity of the brick is 0.5 W/m °C and wall thickness is 200 mm. (L3/CO1) 6
- C) What is the law of heat conduction? State the assumptions on which this law is based? (L1/CO1) 6

Q.2 Solve Any Two of the following.

12

- A) Derive general heat conduction equation in Cartesian co-ordinate system. (L2/CO2) 6
- B) Define Nusselt number, Prandtl number and Grashoff number. (L2/CO2) 6
- C) Write a short note on critical radius of insulation. (L1/CO2) 6

Q. 3 Solve Any Two of the following.

12

- A) Consider the flow of a gas with density 1 kg/m³, viscosity 1.5×10^{-5} kg/(m.s), specific heat $C_p = 846$ J/(kg.K) and thermal conductivity $k = 0.01665$ W/(m.K), in a pipe of diameter $D = 0.01$ m and length $L = 1$ m, and assume the viscosity does not change with temperature. The Nusselt number for a pipe with (L/D) ratio greater than 10 and Reynolds number greater than 20000 is given by “ $Nu = 0.026 Re^{0.8} Pr^{1/3}$ ”. While the Nusselt number for a laminar flow for Reynolds number less than 2100 and $(Re Pr D/L) < 10$ is “ $Nu = 1.86 [Re Pr (D/L)]^{1/3}$ ”.
If the gas flows through the pipe with an average velocity of 0.1 m/s, the heat transfer coefficient is? (L1/CO3) 6
- B) What is convective heat transfer? Distinguish between free and forced Convection (L2/CO4) 6

- C) Explain the concept of hydrodynamic and thermal boundary layers with reference to flow over a flat heated plate. (L3/CO5) 6

Q.4 Solve Any Two of the following. 12

- A) What is heat exchanger and types of heat exchanger (any 05)? (L2/CO5) 6
- B) Derive the expression for LMTD method for an analysis of counter flow heat exchanger. (L3/CO2) 6
- C) Water enters a parallel flow double-pipe heat exchanger at 15 °C, flowing at the rate of 1200 kg/hr. It is heated by oil ($C_p = 2000 \text{ J/kg.K}$), flowing at the rate of 500 kg/hr from an inlet temperature of 90 °C. For an area of 1 m² and an overall heat transfer coefficient of 1,200 W/m².K determine the total heat transfer and the outlet temperatures of water and oil. (L3/CO3) 6

Q. 5 Solve Any Two of the following. 12

- A) Define shape factor. Explain relations/theorems of shape factor. (L2/CO5) 6
- B) What is Stefan Boltzmann Law? Explain the concept of total emissive power of a surface. (L1/CO6) 6
- C) Two very large parallel plates with emissivity 0.5 exchange heat. Determine the percentage reduction in the heat transfer rate if a polished aluminium radiation shield of emissivity = 0.25 is placed in between the plates. (L3/CO6) 6

***** End *****